Worksheet 2.2 — Factoring Polynomials

Grade 11 Functions, University (MCR3U) · Unit 2: Equivalent Algebraic Expressions

The Big Ideas

Factoring is expanding in reverse: it turns a sum of terms into a product. Factors are useful because each factor $(x - a)$ gives an x-intercept of the graph at $x = a$, and because they let us cancel in fractions. **Every factorisation can be checked by expanding it.**

- **Order of attack:** (1) GCF first; (2) count the terms — 2: difference of squares; 3: trinomial; 4: grouping; (3) factor **completely**; (4) check by expanding.
- **Simple trinomial** $x^2 + bx + c$: two numbers with product c and sum b. **Complex trinomial** $ax^2 + bx + c$: product ac, sum b; split the middle term and group.
- **Patterns:** $a^2 - b^2 = (a - b)(a + b)$ and $a^2 \pm 2ab + b^2 = (a \pm b)^2$. A *sum* of squares $a^2 + b^2$ does not factor over the integers.
- **Quadratic form:** $x^4 - 13x^2 + 36$ is a trinomial in $u = x^2$; factor it, then factor again.
- **Graph link:** the factored form shows the x-intercepts at a glance: $y = (x - 5)(x + 3)$ crosses at 5 and -3.

Concept Essentials

Factoring undoes expanding

Factoring writes a sum as a product. Always **check by expanding**. Start with the **GCF**, then count terms: 2 terms → difference of squares; 3 terms → trinomial; 4 terms → grouping. Keep going until no factor can be factored again.

Trinomials

$x^2 + bx + c$: find two numbers with product c and sum b. $ax^2 + bx + c$: find two numbers with product ac and sum b, split the middle term, and group. If no pair works, the trinomial is prime.

Why factors matter

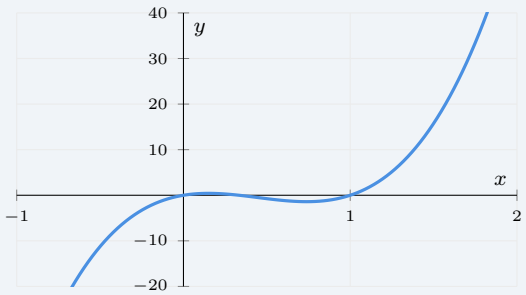
Each factor $(x - a)$ means the graph crosses the x-axis at a. Factors also let you cancel in rational expressions (2.3) and solve equations (the zero-product property).

Worked Examples

Example 1 — Greatest common factor (two variables)

Factor $18x^3y - 24x^2y^2 + 6xy$.

Solution. Numbers: $\gcd(18, 24, 6) = 6$; variables: xy is in every term. GCF $= 6xy$.
Divide each term: $3x^2$, $-4xy$, 1.
$= 6xy(3x^2 - 4xy + 1)$ — note the last term leaves a *1*, not nothing.
Check: at $x = y = 1$: $18 - 24 + 6 = 0$, and $6(3 - 4 + 1) = 0$ ✓

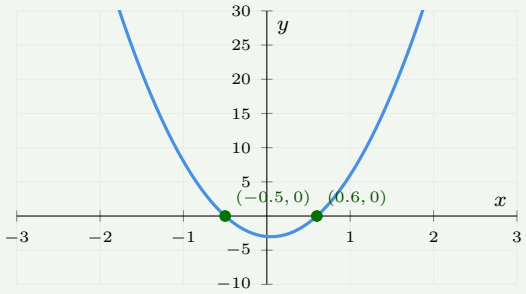


With $y = 1$.

Example 2 — Common binomial factor

Factor $5x(2x + 1) - 3(2x + 1)$.

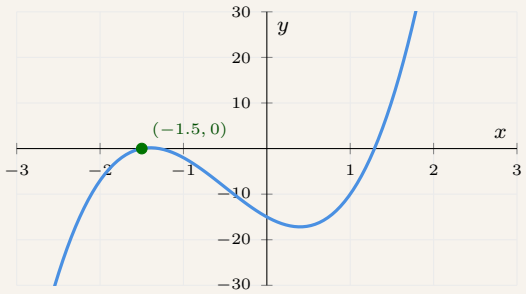
Solution. The bracket $(2x + 1)$ appears in both terms: take it out.
$= (2x + 1)(5x - 3)$ (expanded: $10x^2 - x - 3$).
Check: at $x = 0$: $0 - 3 = -3$, and $(1)(-3) = -3$ ✓



Example 3 — Grouping

Factor $6x^3 + 9x^2 - 10x - 15$.

Solution. Pair the terms: $(6x^3 + 9x^2) + (-10x - 15) = 3x^2(2x + 3) - 5(2x + 3)$.
The brackets match, so take out $(2x + 3)$: $(2x + 3)(3x^2 - 5)$.
Note the -5: taking out -5 from $-10x - 15$ keeps the brackets equal.
Check: at $x = 1$: $6 + 9 - 10 - 15 = -10$, and $(5)(3 - 5) = -10$ ✓



Example 4 — Simple trinomial

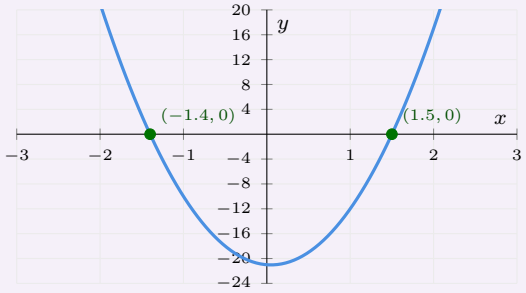
Factor $x^2 - 2x - 48$.

Solution. Product -48, sum -2. Opposite signs; the larger number is negative: -8 and 6.
$= (x - 8)(x + 6)$. Check: $x^2 + 6x - 8x - 48$ ✓.

Example 5 — Complex trinomial by decomposition

Factor $10x^2 - x - 21$.

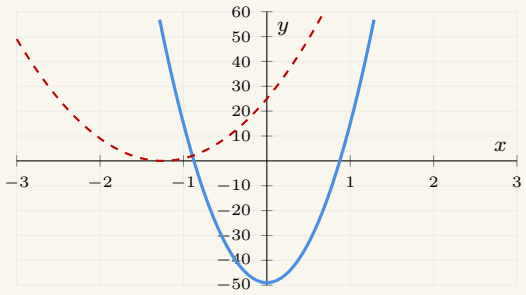
Solution. $ac = -210$; we need product -210 and sum -1: 14 and -15.
$10x^2 + 14x - 15x - 21 = 2x(5x + 7) - 3(5x + 7)$.
$= (5x + 7)(2x - 3)$.
Check: at $x = 1$: $10 - 1 - 21 = -12$, and $(12)(-1) = -12$ ✓



Example 6 — Difference of squares and perfect squares

Factor (a) $64x^2 - 49$ (b) $16x^2 + 40x + 25$.

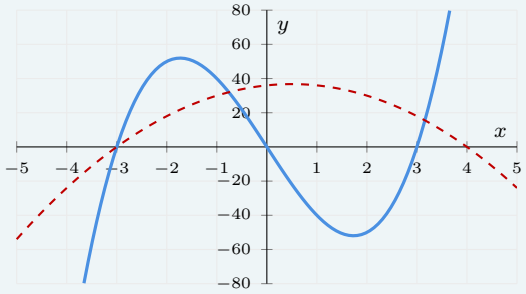
Solution. (a) $(8x)^2 - 7^2 = (8x - 7)(8x + 7)$.
(b) $(4x)^2$, 5^2 and $2(4x)(5) = 40x$ ✓, so $= (4x + 5)^2$.
Check: at $x = 1$: (a) $15 = (1)(15)$; (b) $81 = 9^2$ ✓



Example 7 — Factor completely

Factor (a) $5x^3 - 45x$ (b) $-3x^2 + 3x + 36$.

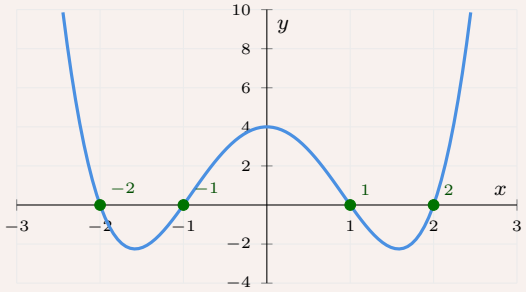
Solution. (a) $5x(x^2 - 9) = 5x(x - 3)(x + 3)$.
(b) Take out -3: $-3(x^2 - x - 12)$; product -12, sum -1: $-4, 3$. $= -3(x - 4)(x + 3)$.
Check: at $x = 1$: (a) $-40 = 5(1)(-2)(4)$; (b) $36 = -3(-3)(4)$ ✓



Example 8 — Quadratic form

Factor $x^4 - 5x^2 + 4$.

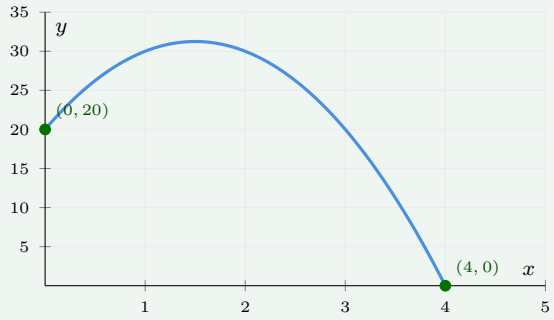
Solution. Let $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4)$.
Back-substitute: $(x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2)$.
Check: at $x = 0$: 4, and $(-1)(1)(-2)(2) = 4$ ✓



Example 9 — Application: when does the ball land?

A ball's height is $h = -5t^2 + 15t + 20$ (metres, t in seconds). When does it hit the ground?

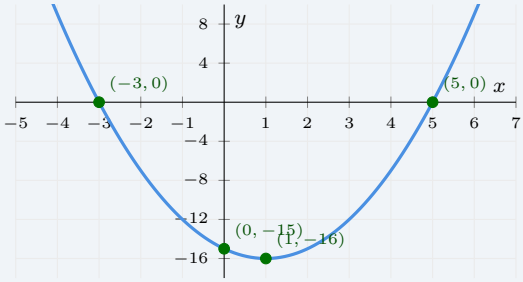
Solution. Factor: $h = -5(t-4)(t+1)$.
$h = 0$ when $t - 4 = 0$ or $t + 1 = 0$: $t = 4$ or $t = -1$. Time cannot be negative, so the ball lands at $t = 4$ s.
The zeros -1 and 4 are symmetric about $t = 1.5$, so the maximum is at $t = 1.5$ s.



Example 10 — Reading a graph from the factored form

Factor $y = x^2 - 2x - 15$ and describe its graph.

Solution. $y = (x-5)(x+3)$: x-intercepts 5 and -3; y-intercept -15; opens up. The vertex lies halfway between the intercepts, at $x = 1$, where $y = -16$.



The factors give the intercepts; symmetry gives the vertex.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Factor $14x^2 - 21x$.

Question 2

Factor $12a^2b^3 - 18ab^2 + 6ab$.

Question 3

Factor completely $x^3 + 5x^2 - 4x - 20$ (use grouping, then look again).

Question 4

Factor (a) $x^2 + 11x + 28$ (b) $x^2 - 3x - 40$. Say why the signs in (b) are opposite.

Question 5

Factor $6x^2 - 17x + 12$ by decomposition. Show the split of the middle term.

Question 6

Factor $12x^2 + 5x - 2$ and check by expanding.

Question 7

Factor (a) $81x^2 - 16$ (b) $25x^2 + 30x + 9$.

Question 8

Factor completely (a) $2x^3 - 18x$ (b) $x^4 - 17x^2 + 16$.

Question 9

Factor $(x - 2)^2 - 36$ and simplify. Then check by expanding the original.

Question 10

A ball's height is $h = -5t^2 + 10t + 15$. Factor, find when it lands, and find its maximum height. (The maximum is halfway between the zeros.)

Question 11

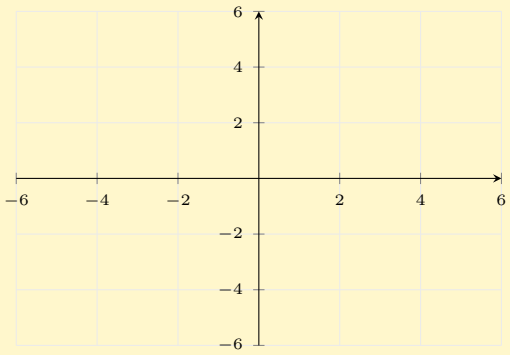
A rectangle has area $x^2 + 14x + 48$ and width $x + 6$. Find its length, then its perimeter when $x = 3$.

Question 12

A student claims that $x^2 + 16 = (x + 4)(x - 4)$. Test the claim with $x = 1$, then explain why $x^2 + 16$ cannot be written as $(x + a)(x + b)$ for integers a, b.

Question 13

Sketch $y = (x+2)(x-1)(x-4)$ by finding its x-intercepts and y-intercept. Does the graph go up or down on the far right? On the far left?



Question 14 — Challenge

Show that $x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1)$. (Hint: write $x^4 + x^2 + 1 = (x^2+1)^2 - x^2$.)

Question Recap & Answer Key

Question Recap

1. Factor $14x^2 - 21x$.
2. Factor $12a^2b^3 - 18ab^2 + 6ab$.
3. Factor completely $x^3 + 5x^2 - 4x - 20$ (use grouping, then look again).
4. Factor (a) $x^2 + 11x + 28$ (b) $x^2 - 3x - 40$. Say why the signs in (b) are opposite.
5. Factor $6x^2 - 17x + 12$ by decomposition. Show the split of the middle term.
6. Factor $12x^2 + 5x - 2$ and check by expanding.
7. Factor (a) $81x^2 - 16$ (b) $25x^2 + 30x + 9$.
8. Factor completely (a) $2x^3 - 18x$ (b) $x^4 - 17x^2 + 16$.
9. Factor $(x - 2)^2 - 36$ and simplify. Then check by expanding the original.
10. A ball's height is $h = -5t^2 + 10t + 15$. Factor, find when it lands, and find its maximum height. (The maximum is halfway between the zeros.)
11. A rectangle has area $x^2 + 14x + 48$ and width $x + 6$. Find its length, then its perimeter when $x = 3$.
12. A student claims that $x^2 + 16 = (x + 4)(x - 4)$. Test the claim with $x = 1$, then explain why $x^2 + 16$ cannot be written as $(x + a)(x + b)$ for integers a, b.
13. Sketch $y = (x + 2)(x - 1)(x - 4)$ by finding its x-intercepts and y-intercept. Does the graph go up or down on the far right? On the far left?
14. Show that $x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1)$. (Hint: write $x^4 + x^2 + 1 = (x^2 + 1)^2 - x^2$.)

✓ Answer Key

1. $7x(2x - 3)$.
2. $6ab(2ab^2 - 3b + 1)$. Check: $6ab \cdot 2ab^2 = 12a^2b^3$ ✓.
3. $x^2(x + 5) - 4(x + 5) = (x + 5)(x^2 - 4) = (x + 5)(x - 2)(x + 2)$.
4. (a) $(x + 4)(x + 7)$ (b) $(x - 8)(x + 5)$. In (b) the product -40 is negative, so one number is positive and one negative; their sum is -3, so the larger in size (-8) is negative.
5. $ac = 72$, sum -17: -8 and -9. $6x^2 - 8x - 9x + 12 = 2x(3x - 4) - 3(3x - 4) = (2x - 3)(3x - 4)$.
6. $ac = -24$, sum 5: 8 and -3. $12x^2 + 8x - 3x - 2 = 4x(3x + 2) - 1(3x + 2) = (4x - 1)(3x + 2)$.
7. (a) $(9x - 4)(9x + 4)$ (b) $(5x + 3)^2$.
8. (a) $2x(x - 3)(x + 3)$ (b) $(x - 1)(x + 1)(x - 4)(x + 4)$.
9. $(x - 2 - 6)(x - 2 + 6) = (x - 8)(x + 4)$. Expanding: $x^2 - 4x + 4 - 36 = x^2 - 4x - 32$ ✓.
10. $h = -5(t - 3)(t + 1)$. Lands at $t = 3$ s (the other zero -1 is rejected). Zeros -1 and 3, so the maximum is at $t = 1$: $h(1) = -5 + 10 + 15 = 20$ m.
11. $x^2 + 14x + 48 = (x + 6)(x + 8)$: the length is $x + 8$. For $x = 3$: 9 by 11, perimeter $2(9 + 11) = 40$.
12. At $x = 1$: $1 + 16 = 17$ but $(1 + 4)(1 - 4) = -15$, so the claim is false. For $(x + a)(x + b) = x^2 + (a + b)x + ab$ we would need $a + b = 0$ and $ab = 16$; but $a + b = 0$ means $b = -a$, so $ab = -a^2 \leq 0$, never 16.
13. Intercepts: $x = -2$, 1, 4; y-intercept: $(2)(-1)(-4) = 8$. The leading term is x^3, so the right end goes *up* and the left end goes *down*.
14. $(x^2 + 1)^2 - x^2 = x^4 + 2x^2 + 1 - x^2 = x^4 + x^2 + 1$, and it is a difference of squares: $(x^2 + 1 - x)(x^2 + 1 + x) = (x^2 + x + 1)(x^2 - x + 1)$.